

Edit or's comment

Dear Dr Moon,

Your manuscript, "Radionuclide Identification for Plastic Scintillation Detectors by Backward Cumulative Channel Sum", has now been assessed.

We invite you to revise your paper, carefully addressing the comments from the reviewers and the editor. Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained. When your revision is ready, please submit the updated manuscript and a point-by-point response. This will help us move to a swift decision.

Please note that if your manuscript uses any custom or bespoke computational tool or code, or reports a new algorithm, tool, software, or a pipeline (even if individual components are not new), the underlying code must be deposited in a recognised DOI-assigning repository (e.g. zenodo) and linked either from Methods or a dedicated Code Availability section.

Editor Comments

"The reviewers have highlighted major issues with the manuscript, specifically lack of quantitative validation, comparison with established methods and missing details regarding calibration, and quantitative evaluation. "
-Guy Garty

We recommend submitting all revisions within the mentioned deadline.

If you need more time, please contact us and include your submission ID.

Kind regards,

Pranjal Waghmare
Assistant Editor
Scientific Reports
[Support contact: srep@nature.com](mailto:srep@nature.com)

Submission ID: 2cf10123-75ae-402b-ba8e-530bd1dd4208

Reviewer1

The manuscript presents a simple approach for radionuclide identification using plastic scintillation detectors, which could be of interest for radiation portal monitoring. However, there are several important aspects that are not clearly described and make it difficult to fully understand and reproduce the results. In particular, more information is needed about the reference spectra, the detector calibration, and how the fitting is evaluated. It would also be helpful if the authors discussed more clearly the limitations of the method and the conditions in which it can be applied.

Attachment(s):

- [Download Reviewer 1 attachment 1](#)

Reviewer2

No extra comments for the review process.

Reviewer3

This manuscript addresses a highly relevant and practically important challenge in radiation detection, namely the reliable identification of radionuclides using plastic scintillation detectors under low signal-to-background conditions. Given the widespread deployment of these detectors in radiation portal monitoring systems due to their cost-effectiveness, robustness, and high efficiency, improving their analytical performance without requiring hardware modifications is of significant interest to both the scientific community and applied security domains. The proposed BCCS-based spectral processing framework offers a promising and conceptually straightforward approach to enhancing spectral interpretability and radionuclide separability, which is critical for real-world monitoring scenarios. Overall, the work is well motivated and targets a problem of clear operational and scientific importance.

The authors provide a clear and candid discussion of the method's limitations, acknowledging dependencies on reference spectra quality, environmental stability, and background representativeness. Their identification of these factors, along with suggested future improvements, reflects a careful and balanced assessment that strengthens the overall credibility of the study.

My main concern with the manuscript is that, although it claims that the proposed method enhances spectral separability and radionuclide identification, it does not provide sufficient quantitative results to support these claims. Furthermore, the manuscript does not include comparisons with alternative methods. In addition, the approaches used to evaluate the reliability of the results appear somewhat heuristic and are not grounded in established statistical frameworks, such as those described in ISO 11929.

For these reasons, I recommend that the authors strengthen the manuscript by comparing the sensitivity of the proposed method with at least two commonly used approaches:

1. Least-squares fitting using the original (untransformed) spectra, where the background spectrum is included as a template rather than subtracted prior to analysis.
2. An energy window method in which the windows correspond to the dominant regions of Am-241, Cs-137, and Co-60, with an additional high-energy window used to estimate the background level.

Minor comments:

The structure of the manuscript is somewhat unconventional. The authors should consider presenting the Methods section before the Results section.

The advantages of the two background compensation methods are described in both the subsections "Transformation of reference radionuclide spectra" and "Background compensation for mixed-source spectra." These should be consolidated, preferably in the Methods section.

Due to the fixed scaling of the y-axis, Figure 4 is difficult to read.

The y-axis of Figure 5 should be labeled.

The measurement times for the analyzed spectra, background spectra, and template spectra are not clearly defined and should be explicitly stated.